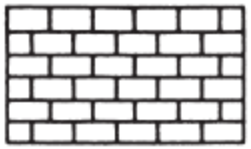
Estimating in Building Construction

Chapter 11 Masonry

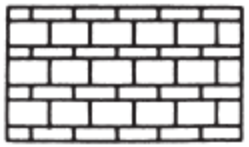
Masonry Includes

- Block
- Brick
- Clay tile
- Stone
- Lintels
- Reinforcing
- Weep holes
- Precast concrete
- Mortar
- Scaffolding
- Temporary enclosures

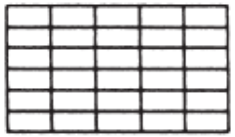
Concrete Block Patterns



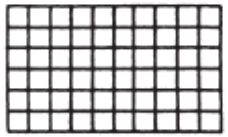
Running bond



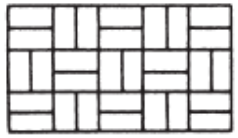
Coursed ashlar (alternate 4" and 8" high units)



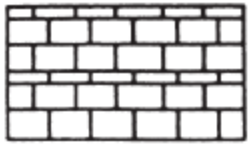
Horizontal stacking



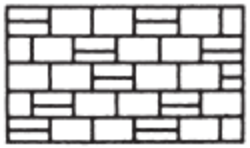
Square stacking



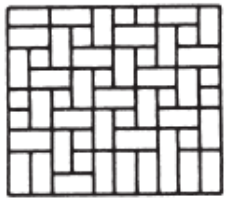
Basket weave



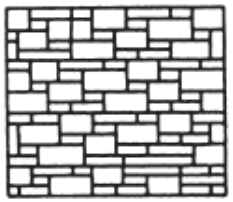
Coursed ashlar (4" and 8" high units)



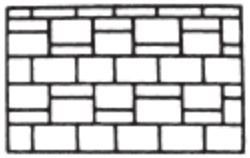
Coursed ashlar (4" and 8")



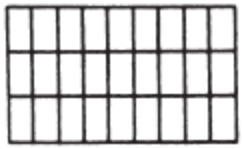
Coursed pattern ashlar (8 x 8 and 8 x 16 units)



Coursed ashlar (4 x 8, 4 x 12, 4 x 16, 8 x 12 and 8 x 16 units)

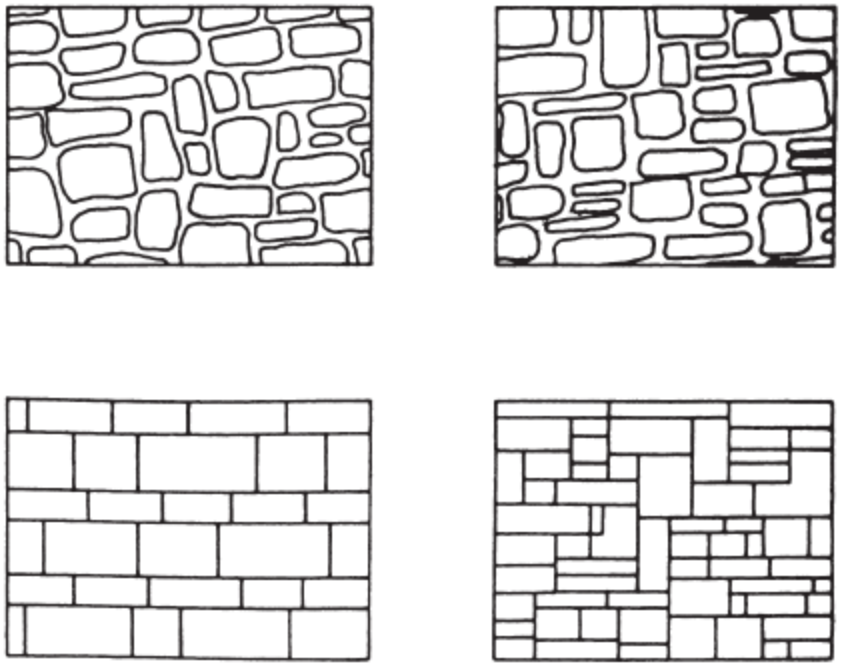


Coursed ashlar (4" and 8")

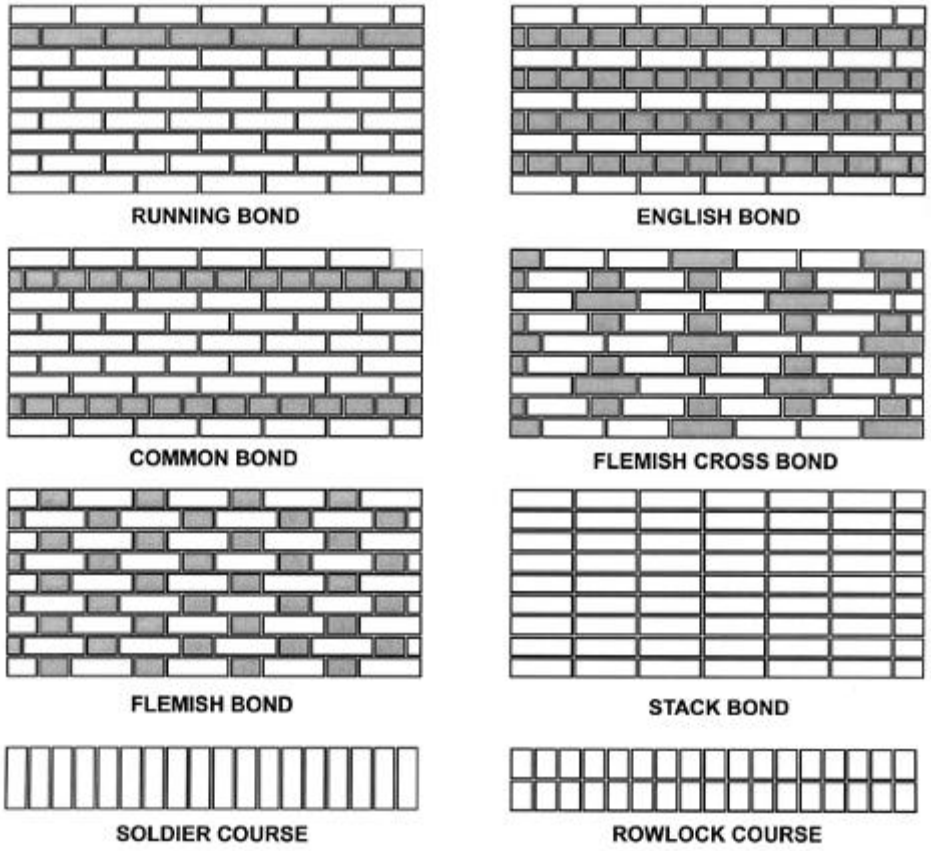


Vertical stacking (8")

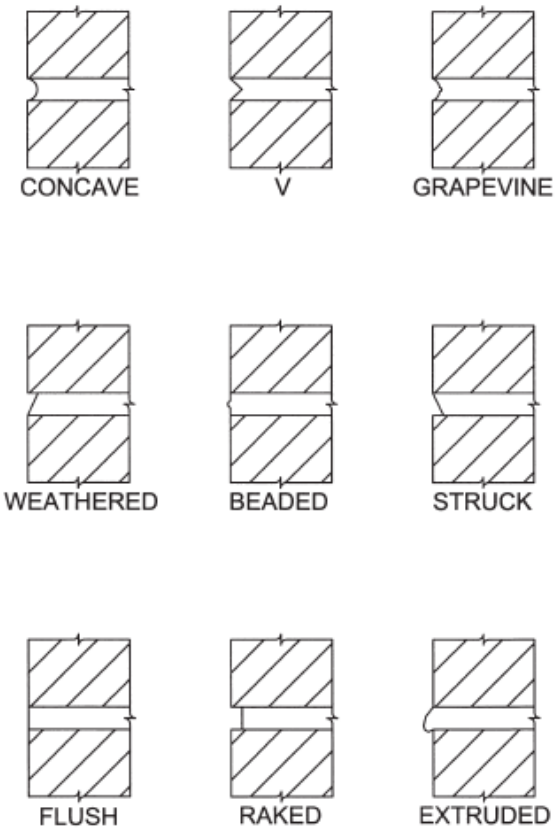
Stone Patterns



Brick Patterns



Mortar Joints



Concrete Masonry

- Comprises all molded concrete units e.g. concrete brick, hollow and solid blocks, decorative blocks

Concrete Block

- Materials bid by the block
  - Keep block types separate, heavyweight or lightweight

Say 112 Units per 100 square foot of wall

For nominal size 8″ x 8″ x 16″ block

Units per sq. ft. = 144/(8″ x 16″)

= 1.125 block per sf

Concrete Block

- Labor bid by the square foot
- Strength and absorption (ASTM)
- Corners should be taken once
- Deductions for all opening in excess of 10 sf.
- Add allowances for waste and breakage

Separate Masonry By:

- Size
- Shape
- Colors
- Bond or pattern
- Shape of mortar joints
- Mortar color
- Other (e.g. fire rating)

Material Requirements

Materials	Materials (per 100 s.f. face area			
	Concrete Block			Brick
	8 x 16	8 x 18	5 x 12	2.67 x 4 x 8
No. of Units	112.5	100	240	675
Mortar (Cu. Ft.) Face shell bedding	2.3	2.2	3.6	7.2
Mortar (Cu. Ft.) Face shell and web bedding	3.2	3.0	2.5	7.2

Add waste for blocks and bricks (5 to 100%)

Block Productivity

Labor Hours per 100 Square Feet				
Wall Thickness	6″		8″	
Type of Work	Mason	Laborer	Mason	Laborer
Simple foundation			4.5 – 6.0	4.0 – 7.5
Foundation with several corners and openings			5.0 – 7.5	5.0 – 7.5
Exterior walls to 4′ high	4.0 – 6.0	4.0 – 6.5	4.5 – 6.0	5.0 – 7.5
Exterior walls 4′ – 8′	4.0 – 6.5	4.5 – 7.0	4.5 – 6.5	6.0 – 9.0
Exterior walls above 8′	5.0 – 9.0	7.0 – 10.0	5.0 – 7.0	7.0 – 10.0
Interior partitions	3.5 – 6.5	4.5 – 7.5	4.5 – 6.0	5.0 – 7.5

Clay Masonry

- Includes brick and tiles
- Brick is considered a solid unit if it contains cores less than 25% of the cross-sectional area of the unit.
- Facing tiles are available either glazed or unglazed
- Ceramic glazed units are available in wide variety. They are easy-to-clean

Bricks

- Brick is sold by the thousand units
- Bid by the 1,000 bricks
- Labor cost is affected by length of straight wall, windows, piers or anything else that might slow the mason’s work such as weather.

Brick Productivity

Labor Hours per 1,000 Standard Size or Modular Bricks		
Type of Bond	Mason	Laborer
Common	10.0 to 15.0	11.0 to 15.5
Running	8.0 to 12.0	9.0 to 13.0
Stack	12.0 to 18.0	10.0 to 15.0
Flemish	11.0 to 16.0	12.0 to 16.0
English	11.0 to 16.0	12.0 to 16.0

Tiles

- Calculate area and then number of units or tiles

Stone Masonry

- Used for:
  - Exterior walls
  - Walkways
  - Riprap
  - Veneer for walls
- Include
  - Mortar (cu. ft.)
  - Ties
  - Flashing (sq. ft.)
  - Lintels (lf.)

Estimated by the area in sq. ft. with the thickness given
Cut stones have 2 to 4 % of the total volume as mortar

Stone Masonry

- Dressing and cutting of stone require skill on the part of the mason
- Large pieces may require crane for lifting

Mortar

- Made from:
  - Portland cement
  - Mortar sand
  - Hydrated lime
- Thickness of mortar joints 3/8”, ¼”, ½”
- Use two mixers when using two colors
- Labor cost is included as part of the masonry units

Accessories

- Masonry wall reinforcing comes in a wide variety of styles and wire gauges (see Fig. 11.15, Fig. 11.16)
- Control joints
  - A vertical joint from top to bottom. Reinforcing should not cross over control joint
- Wall ties
  - Connect inner and outer wythes (Fig 11.17)
- Flashing
  - Diverts moisture to the exterior. Materials include copper, aluminum, copper and polyethylene, plastic sheeting, elastomeric compound, copper and fabric (Fig. 11.18)

Accessories

- Weep holes
  - Drain moisture from the wall
- Lintels
  - Horizontal member that supports masonry
- Sills
  - Members at the bottom of the windows

Accessories

- Coping
  - Cover top course of wall (see Fig. 11.18)

Labor Hours for Installing Accessories

Accessories	Unit	Labor Hours
Wall Reinforcing	1,000 l.f.	1.0 to 1.5
Lintels – Precast Concrete (up to 300 pounds)	100 l.f.	4.5 to 7.0
Coping, Sills, and Precast Concrete (up to 150 pounds)	100 l.f.	5.0 to 8.0
Wall Ties	100 Pieces	1.0 to 1.5

Reinforcing

- Rebar
  - Placed in masonry walls, beam block and grouted
  - Bid same as rebar in concrete

Equipment



- Mason's hand tools
- Mortar boxes
- Mortar mixer
- Hoses
- Shovels
- Wheelbarrows
- Mortar tubs
- Scaffolding
- Hoisting equipment
- Forklifts
- Lift trucks



Cold Weather

- Temporary enclosure
- Temporary heat
- Common practice to heat the water to maintain temperature

Thank You